

2023



MUFG Accelerates Business Transformation with a Modern Data Architecture Consolidating over 40M Customer Accounts

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Approximately 40 million

Number of customer accounts stored in the data lake

Approximately 500TB

Volume of data in the data lake

80TB

Volume of data in the data warehouse

Over double

Increase in size of the in the data lake and data warehouse in 1 year



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Overview

Japan's largest bank, [MUFG Bank, Ltd.](#) consolidated data from over 40 million customers into an Amazon S3 Data Lake and Amazon Redshift Data Warehouse to provide better consistency and access for business leaders at all levels to make higher quality and faster business decisions. This reduced data processing work across the business to enable frontline managers to concentrate on branch sales and operations, where the introduction of new Business Intelligence (BI) tools also provided senior executives with a clearer view of business operations through more specific and insightful management reports.

MUFG has digital transformation as a core pillar of its FY2021-2023 Medium-Term Business Plan and the adoption of Amazon Web Services (AWS) for its modern data architecture was one of the first key steps to become a "digitized financial platform" that combines digital convenience and innovation with safety and security.



Opportunity | Integrating Data Across Analytical Systems In An On-premises Environment

MUFG captures its purpose with the phrase "Committed to empowering a brighter future.," where 'Digital' is one of the society-altering sea changes that they are addressing. MUFG has digital transformation as core pillar of its FY2021-2023 Medium-Term Business Plan and the adoption of Amazon Web Services (AWS) for their modern data architecture was one of the first key steps to become a "digitized financial platform" that combines digital convenience and innovation with safety and security.

Mr. Yuji Fujisaku, Managing Director at the Corporate Infrastructure Transformation Office in the Corporate Planning Division, explains: "Collecting and storing data that is not utilized is a cost. We thought about using current data first while continuing to perform data collection and maintenance. We decided to create an environment where all bank employees from bank headquarters to branch offices can obtain data that will help them to make quick decisions."



MUFG first considered building a data utilization platform in 2015 and began to build in earnest when digital transformation became a core pillar of their strategy; the Group decided to begin by introducing this platform at MUFG Bank.

In 2018, there were close to 20 analysis/reporting systems within MUFG Bank, such as business management systems and risk management systems," recalls Toru Kuwajima, Vice President at the Corporate Infrastructure Transformation Office in the Corporate Planning Division. "Since many of these systems were operated in an on-premises environment, it was necessary to collect data from different individual systems and the available data was limited to going back just a few years."

To solve this issue, MUFG began building a data utilization platform consisting of an [Amazon Simple Storage Service](#) (Amazon S3) data lake that stores all kinds of in-house data, including structured and unstructured data, an Amazon Redshift data warehouse that stores processed data, and BI tools that visualize data.

Solution | Building A Group-wide Data Platform For Structured And Unstructured Data

For the basis of the infrastructure, MUFG had decided to use AWS, having stated that it would make full-scale adoption of AWS at the beginning of 2017. However, the project team had to first reach an agreement with the Information Security Department to address storing data on the cloud. Mr. Satoshi Hirabayashi of the Data Strategy Systems Department at Mitsubishi UFJ Information Technology (MUIT), which is the Group's IT company, says, "We were able to obtain agreement by identifying external risks such as cyberattacks and internal risks such as unauthorized access, and by explaining the safety of data utilization on the cloud after thoroughly examining security implementation on AWS." Mr. Toru Matsushita, who also works on data strategy at MUIT, recalls: "As a result, the security rules we put in place and the functions that were implemented became security standards across MUFG Bank when utilizing the cloud, and this formed the basis for subsequent projects to proceed smoothly."

The solution was to construct a modern data architecture in Oct 2019, consisting of an Amazon S3 data lake and an [Amazon Redshift](#) data warehouse. BI tools were released for headquarters in October 2020. In December 2022, BI tools were also rolled out at branch offices, so that frontline teams could use data through BI



“We have gradually enhanced the functionality. In addition to adopting [Amazon Kinesis](#) for real-time data collection in 2022, we are currently verifying the [AWS Glue](#) ETL tool. In the future, we plan to evolve this with [tools such as] Amazon Redshift Serverless and Redshift Data Sharing.” (Mr. Matsushita)

Currently, the data utilization platform stores structured data such as information from 40 million customer accounts and transaction statements from the past 10 years, as well as unstructured data such as image data of account transfer request forms. As of December 2022, the amount of data was approximately 500TB in the data lake and 80TB in the data warehouse.

“The amount of data over the past year has more than doubled, with the data lake growing by about 300TB and the data warehouse growing by about 40TB. Since there is no point in storing a large amount of data if it cannot be used, an intranet portal has been constructed to make it easier to search data and metadata and then extract results. On the portal, users can search for data, download it, or transfer it to the Amazon Redshift data warehouse depending on their granted permissions. The portal site also plays the role of a data hub for various systems within the company.” (Mr. Hirabayashi)

Outcome | Acquiring Data For Faster Decision-making And Enabling The Shift To Data-driven Management

With a modern data architecture, MUFG Bank has realized productivity gains in branches and better-quality management insights in the bank’s head office.

“As a result of making it possible to cross-search data for all branches and all customers, it is now also possible to extract corporate data from deposit/withdrawal statements for the past year and analyze them by industry. In the middle of the COVID-19 pandemic, a report analyzing year-on-year performance by industry was prepared, and this was highly praised by management, which further accelerated the trend toward data-driven management.” (Mr. Kuwajima)

Previously, data collected from headquarters was processed at each branch to analyze customer profiles and year-on-year performance, etc., but by providing dashboards for branch offices, it is now possible to reduce data processing/aggregation work and concentrate on sales operations.

“Small branch offices were burdened by processing and aggregating data returned from headquarters. We have worked with the headquarters departments that support sales, and gradually they have shifted to returning information using the dashboard. I have heard that branch managers, section chiefs, and personnel are capturing the same data, such as the progress of KPIs, at the same time and with the same granularity, and rational and quick decisions are beginning to be made. This is just the beginning, but we have already been able to take a step towards data-driven management, which is the goal for all bank personnel.” (Mr. Fujisaku)

This has also contributed to the enhancement of bank headquarters operations. The company is applying it in various ways, from product development to sales support measures, cost analysis and profit analysis, and provision as a sales support dashboard by adopting a new approach to risk management data.

“I feel that the preparatory phase is finally over. Going forward, our aim is to ensure that we have a deeply rooted culture of data utilization by accumulating many successful experiences, even if they are small, from the management team to the headquarters and branch offices.” (Mr. Fujisaku)

To promote data-driven management, MUFG is also expanding available BI training and qualification programs in parallel. In the next stage, they plan to establish the foundations to enable AI-generated signals and automated



actions.

“Instead of responding to requests from headquarters and branch offices, technical departments and IT departments can spend more time to anticipate needs, develop new services, and actively roll them out to the users. In terms of our future outlook, we are planning horizontal expansion to Group companies while expanding the range of data to be collected.” (Mr. Fujisaku)

Company Overview of MUFG Bank, Ltd.

MUFG Bank is Japan's largest bank with over 360 years' history. As part of the Mitsubishi UFJ Financial Group, they are present across Japan and in approximately 50 other countries. As part of their FY2021-2023 medium-term management plan, the Bank is undergoing “3 years of new challenges and transformation” based on the 3 pillars of Corporate Transformation, Strategies for Growth, and Structural Reforms. For their digital transformation, the Bank is expanding the functions of online channels and accelerating their shift to becoming a financial and digital platform operator that combines digital convenience and innovation with safety and security.





Mr. Yuji Fujisaku



Mr. Toru Kuwajima



Key Services Currently In Use

Mr. Toru Matsushita

Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance. S3 is built to store and retrieve any amount of data from anywhere.

[Learn more »](#)

Amazon Redshift

Amazon Redshift uses SQL to analyze structured and semi-structured data across data warehouses, operational databases, and data lakes, using AWS-designed hardware and machine learning to deliver the best price performance at any scale.



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Amazon Kinesis

Amazon Kinesis makes it easy to collect, process, and analyze streaming data in real time so it can be gained timely insights and respond quickly to new information.

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AWS Glue

AWS Glue is a serverless data integration service that makes it easier to discover, prepare, move, and integrate data from multiple sources for analytics, machine learning (ML), and application development.

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Mr. Satoshi Hirabayashi

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